Human body analysis

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Interface description

Identify the gesture type in the picture, return the gesture name, gesture rectangular box, probability score, and can identify 24 common gestures, which are suitable for gesture special effects, smart home gesture interaction and other scenarios.

List of 24 supported gestures: fist, OK, prayer, bow, farewell, one-handed heart, like, Diss, I love you, palm up, two-handed heart (3 types), number (9 types), Rock, vertical middle finger.

It is mainly suitable for selfies within 3 meters and other people's shooting. The identification effect within 1 meter is the best. When the shooting distance is too long, the hand target is too small to accurately locate and identify.

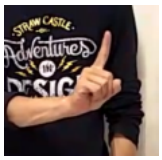
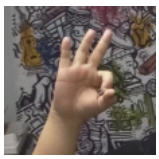
When there are multiple gestures in the picture, it can also be recognized, but in this case, the target of a single gesture may be smaller and the angle may not be good (such as tilting, occlusion, etc.), and the recognition effect may be affected. It is recommended to identify a single gesture, which is the best effect.

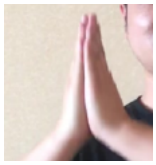
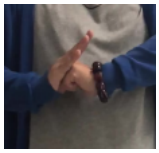
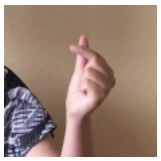
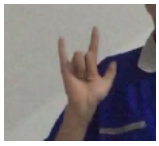
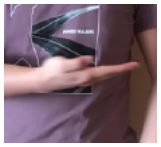
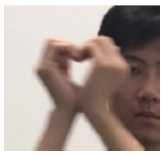
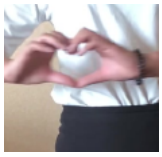
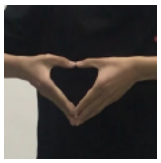
Note:

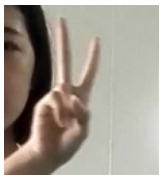
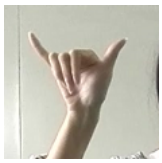
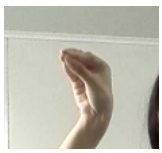
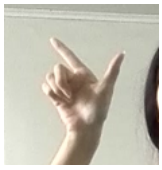
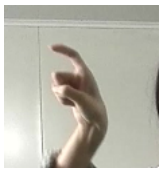
1) Other gestures other than the above 24 categories will be divided into other categories.

2) In addition to identifying gestures, if a face is detected in the image, it will return to the position of the face frame at the same time.

The diagram of the 24 recognizable gestures is as follows:

Serial number	Gesture name	class name	Sample picture
1	Number 1 (original index finger)	one	
2	Number 5 (original palm forward)	five	
3	fist	fist	
four	OK	OK	

5	Pray	Prayer	
6	bow	Congratulations	
7	Fadewell	Honour	
8	One-handed comparison	Heart_single	
9	Like	thumb_up	
10	Diss	thumb_down	
11	I love you.	ILY	
12	Palm up	Palm_up	
13	Hands compare heart 1	Heart_one	
14	Hands compare heart 2	Heart_2	
15	Hands compare heart 3	Heart_3	

16	Number 2	two	
17	Number 3	three	
18	Number 4	four	
19	Number 6	six	
20	Number 7	seven	
21	Number 8	eight	
22	The number 9	nine	
23	rock	rock	
24	Vertical middle finger	Insult	

Online debugging

You can debug the interface in the sample code center, which can perform signature verification, view the request content and return results of online calls, and automatically generate sample code.

Request for explanation

Request example

HTTP method: `POST`

Request URL: `https://aip.baidubce.com/rest/2.0/image-classify/v1/gesture`

URL parameters:

Parameters	Value
access_token	Access_token obtained through API Key and Secret Key, please refer to "Access Token Acquisition"

The header is as follows:

Parameters	Value
Content-type	application/x-www-form-urlencoded

Place the request parameters in the Body, and the details of the parameters are as follows:

Request parameters

Parameters	Is it necessary to choose?	Type	Optional value range	Explain
image	Yes	string	-	Image data, urlencode after base64 encoding, requires that the size not exceed 4M after base64 encoding and urlencode. The base64 encoding of the picture does not include the picture header, such as (``), supports picture formats: jpg, bmp, png, the shortest side is at least 50px, the longest side is up to 4096px

Sample request code

Tip 1: Before using the sample code, please remember to replace the sample Token, picture address or Base64 information.

Tip 2: For classes or libraries that some languages depend on, please check the download address in the code comments.

bash PHP Java Python Cpp C#

```
手势识别
curl -i -k 'https://aip.baidubce.com/rest/2.0/image-classify/v1/gesture?access_toke
```

Return to the description

Return parameters

Field	Is it necessary to choose?	Type	Explain
result_number	Yes	Int	The number of results
result	Yes	object[]	Detected targets, gestures, faces
+class name	No	string	Target category, 24 kinds of gestures, other, face
+top	No	Int	Coordinates on the target box
+width	No	Int	The width of the target box
+left	No	Int	The leftmost coordinate of the target box
+height	No	Int	The height of the target box
+probability	No	float	The probability that the target belongs to this category
log_id	Yes	Int64	The unique log id is used for problem positioning.

Return to the example

```
{
    "log_id": 4466502370458351471,
    "result_num": 2,
    "result": [{
        "probability": 0.9844077229499817,
        "top": 20,
        "height": 156,
        "classname": "Face",
        "width": 116,
        "left": 173
    },
    {
        "probability": 0.4679304957389832,
        "top": 157,
        "height": 106,
        "classname": "Heart_2",
        "width": 177,
        "left": 183
    }]
}
```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